

SECTION A (25 marks)

This section consists of **3** questions. Answer **all** questions.

1. Evaluate the following limits, if exists:

(a) $\lim_{x \rightarrow -\infty} \frac{3x}{\sqrt{x^2 - 36}}$. [4 marks]

(b) $\lim_{x \rightarrow 3} \frac{\sqrt{x^2 + 7} - 4}{x^2 - 3x}$. [5 marks]

2. (a) Find the derivative of $f(x) = \sqrt{8x - 3}$ by using the first principle of differentiation. [4 marks]

(b) Find $\frac{dy}{dx}$ for the following:

(i) $y = \sin(\ln 3x)$ [3 marks]

(ii) $y = \frac{x}{x^3 + 1} + e^{2x} \tan x$. [4 marks]

3. Find the inflection point of the function $f(x) = x^3 + 3x^2 - 24x + 20$. [5 marks]

SECTION B (75 marks)

This section consists of 7 questions. Answer **all** questions.

1. Given $z_1 = 3 + 4i$ and $z_2 = 5 - 2i$. Find z_3 in $a + bi$ form if $\frac{1}{z_3} = \frac{1}{z_1} + \frac{1}{z_1 z_2}$.

Hence, express $\overline{z_3}$ in polar form.

[9 marks]

2. (a) Solve the equality $|3x + 1| = |2x - 7|$.

[4 marks]

- (b) Obtain solution set for the following inequalities

(i) $|2x + 7| + 1 \geq 6x$.

[4 marks]

(ii) $\left| \frac{2x + 1}{x - 3} \right| \geq 2$.

[4 marks]

3. (a) A function $f(x)$ is defined as

$$f(x) = \frac{3 - \ln(4 - x)}{2}, \quad x \in \mathbb{R} \text{ and } x < 4.$$

- (i) Determine the inverse function, $f^{-1}(x)$.

[3 marks]

- (ii) Determine whether $f^{-1}(x)$ is one to one.

[3 marks]

- (b) Given the function $g(x) = e^{x+1}$. Find the function $h(x)$ if $(h \circ g)(x) = x + 1$.

[6 marks]

4. A function f is defined as

$$f(x) = -(x - 2)^2 + 1, \quad x \geq 2$$

Determine the inverse function of f .

Hence, sketch the graph of f and f^{-1} on the same axes.

[6 marks]

5. (a) Find the vertical and horizontal asymptotes of

$$f(x) = \frac{2x^2 + 1}{4x^2 - 25} \quad [6 \text{ marks}]$$

- (b) Given

$$f(x) = \begin{cases} \frac{x^2 - 16}{x^2 - 4x} & , \quad x \neq 4 \\ p & , \quad x = 4 \end{cases}$$

where p is a constant.

- (i) Determine $\lim_{x \rightarrow 4} f(x)$ if exists. [3 marks]

- (ii) Find the value of p if $f(x)$ is continuous at $x = 4$. [2 marks]

6. (a) The parametric equations of a curve is given by $x = -9\sin^3 t$ and $y = 3 - 3\cos^3 t$.

- (i) Show that $\frac{dy}{dx} = -\frac{1}{3} \cot t$. By using $\cot t = \frac{\cos t}{\sin t}$, solve $\frac{dy}{dx} = 0$ for $0 \leq t \leq \pi$. [8 marks]

- (ii) Show that $\frac{d^2y}{dx^2} = -\frac{1}{81\sin^4 t \cdot \cos t}$. [4 marks]

- (b) Given $e^{-2y} + 2xy + \ln(1 - 2x) = 2$. Calculate the value of $\frac{dy}{dx}$ when $x = 2$ and $y = 0$. [4 marks]

7. (a) Given a water tank which has the shape of an inverted circular cone water tank with a base radius of 3 meters and a height of 5 meters. If water is being pumped into the tank at the rate of $2 \text{ m}^3/\text{minute}$, find the rate of change in water level rising when the water is 3 meters deep. [5 marks]

- (b) Air is pumped into a spherical balloon at the rate of $12\pi \text{ cm}^3 \text{ s}^{-1}$. Find the rate of increase of the surface area when its radius is 6 cm. [4 marks]

END OF QUESTION

Final Answer:

SECTION A

1. (a) -3 (b) $\frac{1}{4}$
2. (a) $\frac{4}{\sqrt{8x-3}}$ (b) (i) $\frac{\cos(\ln 3x)}{x}$ (b) (ii) $\frac{1-2x^3}{(x^3+1)^2} + e^{2x}(\sec^2 x + 2 \tan x)$
3. Point of inflection = $(-1, 46)$

SECTION B

1. $z_3 = \frac{11}{4} + \frac{13}{4}i$; $\bar{z}_3 = 4.26(\cos(-0.8685) + i \sin(-0.8685))$
2. (a) $x = \frac{6}{5}$, $x = -8$
(b) (i) $\{x : x \leq 2\}$ (ii) $\left\{x : x \geq \frac{5}{4}, x \neq 3\right\}$
3. (a) (i) $f^{-1}(x) = 4 - e^{3-2x}$ (ii) $f^{-1}(x)$ is one to one
(b) $h(x) = \ln x$
4. $f^{-1}(x) = 2 + \sqrt{1-x}$, $x \leq 1$, D.I.Y
5. (a) Vertical Asymptote: $x = \frac{5}{2}$, $x = -\frac{5}{2}$, Horizontal Asymptote : $y = \frac{1}{2}$
(b) (i) $\lim_{x \rightarrow 4} f(x) = 2$ (ii) $p = 2$
6. (a) (i) $\frac{\pi}{2}$ (ii) D.I.Y (b) $\frac{dy}{dx} = -\frac{1}{3}$
7. (a) $\frac{50}{81\pi} \text{ cm s}^{-1}$ (b) $4\pi \text{ cm}^2 \text{ s}^{-1}$

SECTION A (25 marks)

This section consists of 3 questions. Answer **all** questions.

1. Evaluate the following limits:

(a) $\lim_{x \rightarrow -1} \frac{x^3 + 1}{x + 1}$ [3 marks]

(b) $\lim_{x \rightarrow 3} \frac{\sqrt{x+1} - 2}{x - 3}$ [3 marks]

(c) $\lim_{x \rightarrow \infty} \frac{(2-x)(x-1)}{(x-3)^2}$ [3 marks]

2. (a) Find $\frac{dy}{dx}$ if:

(i) $y = \ln[\sin(9 - 2x)]$ [4 marks]

(ii) $y = \tan(e^{2x})$ [3 marks]

(b) Evaluate $\frac{dy}{dx}$ for $y = (x^2 + 1)^5$ when $x = -1$. [4 marks]

3. Given a function $f(x) = 2x^4 - 16x^2 + 14$. By using the second derivative test, find the maximum and the minimum points of $f(x)$. [5 marks]

SECTION B (75 marks)

This section consists of 7 questions. Answer **all** questions.

1. Given $z_1 = 2 + 3i$ and $z_2 = 4 - 4i$. Express $z = \frac{(z_2)}{(z_1)} + \left[\left(\frac{i^3}{-z_2} \right) \right]$ in Cartesian form.

Hence, find the modulus, argument and polar form for z .

[10 marks]

2. Find the solution set of the inequality $\frac{1}{|x-1|} \leq \frac{2}{|x+2|}$.

[10 marks]

3. If given $g(x) = \frac{x+a}{bx+c}$ and $g^{-1}(x) = 1 + \frac{3}{x+1}$, find the values of the constants a , b and c .

[6 marks]

4. The function f and g are defined as

$$f(x) = 3\sqrt{1-x}$$

$$g(x) = \ln\left(\frac{x-2}{3}\right)$$

- (a) If the function $h(x)$ is defined as $h(x) = \begin{cases} f(x) \\ g(x) \end{cases}$ for $x \in \mathbb{R}$, sketch $h(x)$ and state the domain and range of $h(x)$.

[4 marks]

- (b) Determine whether $f(x)$ is a one to one function.

[2 marks]

- (c) Find $g^{-1}(x)$ and hence, sketch $g^{-1}(x)$ on the same axes with $g(x)$. Hence, state the domain and range of $g^{-1}(x)$.

[6 marks]

5. Consider the function:

$$f(x) = \begin{cases} \frac{x}{x-3} & , \quad x < -3 \\ x+1 & , \quad -3 \leq x \leq 3 \\ (kx-3)^2 & , \quad x > 3 \end{cases}$$

where $k > 1$.

- (a) Determine whether $f(x)$ is differentiable at $x = -3$. [5 marks]
- (b) Determine the value of k if the function $f(x)$ is continuous at $x = 3$. [4 marks]
- (c) Find the horizontal asymptote of the graph of $f(x)$ as $x \rightarrow -\infty$. [2 marks]

6. (a) Parametric equation of a curve is given by $x = t^3 - 3t$ and $y = 3t^2$.
Find

- (i) $\frac{dy}{dx}$ in terms of t . [4 marks]
- (ii) $\frac{d^2y}{dx^2}$ when $t = \sqrt{3}$. [4 marks]

- (b) Find the values of $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ at the point $(1, 3)$ on the curve $3x^2 + y^2 = 4y$. [8 marks]

7. Salt water is poured into a right inverted cone with a semi-vertical angle of 45° at a rate of $3\pi \text{ cm}^3 \text{ s}^{-1}$. Find

- (a) The depth of the salt water after 45 seconds. [7 marks]
- (b) The rate of change of its depth at that instant. [3 marks]

END OF QUESTION

Final Answer:

SECTION A

1. (a) 3 (b) $\frac{1}{4}$ (c) -1
2. (a) (i) $-2\cot(9-2x)$ (ii) $2e^{2x}\sec^2(e^{2x})$ (b) -160
3. (0,14) maximum point ; (2,-18), (-2,-18) minimum points

SECTION B

1. $z = \frac{147}{104} + \frac{45}{104}i$; $|z| = \sqrt{\frac{909}{416}}$; $\text{Arg}(z) = 0.2971\text{rad}$;
 $z = \sqrt{\frac{909}{416}} [\cos(0.2971 \text{ rad}) + i \sin(0.2971 \text{ rad})]$
2. $(-\infty, 0] \cup [4, \infty)$, $x \neq -2$
3. $a = -4$, $b = -1$, $c = 1$
4. (a) $D_h = (-\infty, 1] \cup (2, \infty)$; $R_h = (-\infty, \infty)$
(b) one to one function
(c) $g^{-1}(x) = 3e^x + 2$; $D_{g^{-1}} = (-\infty, \infty)$; $R_{g^{-1}} = (2, \infty)$
5. (a) $f(x)$ is not differentiable at $x = -3$.
(b) $k = \frac{5}{3}$ (c) $y = 1$
6. (a) (i) $\frac{dy}{dx} = \frac{2t}{t^2-1}$; $\frac{d^2y}{dx^2} = \frac{-2-2t^2}{3(t^2-1)^3}$; $\frac{d^2y}{dx^2}\bigg|_{t=1} = -\frac{1}{3}$
(b) $\frac{dy}{dx} = \frac{3x}{2-y}$; $\frac{dy}{dx}\bigg|_{(1,3)} = -3$; $\frac{d^2y}{dx^2}\bigg|_{(1,3)} = -12$
7. (a) $h = 7.3986 \text{ cm}$
(b) $\frac{dh}{dt}\bigg|_{h=7.3986} = 0.0548 \text{ cm/s}$

SECTION A (25 marks)

This section consists of 3 questions. Answer **all** questions.

1. Evaluate each of the following limits, if exists.

(a) $\lim_{x \rightarrow 3} \frac{2x - 6}{-x^2 + 2x + 3}$ [3 marks]

(b) $\lim_{x \rightarrow 9} \frac{x - 9}{3(\sqrt{x} - 3)}$ [3 marks]

(c) $\lim_{x \rightarrow -\infty} \frac{x + 2}{\sqrt{4x^2 + 1}}$ [3 marks]

2. Differentiate the following with respect to x . Give your answer in the simplest form.

(a) $y = \tan^3(e^x + 3)$ [3 marks]

(b) $y = \ln\left(\frac{2 - x^2}{x(1 - x)}\right)$ [4 marks]

(c) $y = 2^{x^3 \ln x}$ [4 marks]

3. The curve $y = 2x^3 + bx^2 + cx$ has a turning point at $(1, 5)$.

(a) Find the values of b and c . [3 marks]

(b) By using the second derivative test, determine whether the stationary point at $(1, 5)$ is a minimum or maximum point.

[2 marks]

SECTION B (75 marks)

This section consists of 7 questions. Answer **all** questions.

1. (a) Given $x = 3 + 2i$, $y = -i$ and $\frac{z}{13} + x = \frac{1}{x} + y^2$. Express z in polar form. [6 marks]

- (b) Given that the complex number, express $z = \sqrt{-1}(2i - i^{10})$ in Cartesian form. [3 marks]

2. (a) Determine the possible values of x if $\left| \frac{2x}{3+x} + 1 \right| \geq 2$.
Express your answer in interval form. [6 marks]

- (b) Find the value of x such that $2|x| \geq 3x - 10$. [5 marks]

3. Given $f(x) = 2x + 1$ and $g(x) = \frac{-3x}{x+1}$.

- (a) State the domain of $f(x)$ and $g(x)$. [2 marks]

- (b) Obtain $f \circ g(k)$ and $g \circ f(k)$. Hence, determine the constant value of k where $f \circ g(k) = g \circ f(k)$. [8 marks]

4. Given

$$f(x) = \begin{cases} -|x+2| & , \quad -5 \leq x \leq -1 \\ 3 & , \quad -1 < x < 1 \\ (x-2)^2 + 1 & , \quad x \geq 1 \end{cases}$$

- (a) Evaluate $f(-3)$. [1 mark]
(b) Sketch the graph of $f(x)$. [3 marks]
(c) Based on the graph, determine the point of x – intercept and y – intercept. [2 marks]
(d) Hence, find the domain and range of f . [2 marks]

5. Given

$$f(x) = \begin{cases} ax+7, & x < -k \\ 3, & -k \leq x < 1 \\ x^2+1, & 1 \leq x < k \\ 3x+1, & k < x < 5 \\ 16, & x \geq 5 \end{cases}$$

(a) Find the values of k if $\lim_{x \rightarrow k} f(x)$ exist. [5 marks]

(b) Hence, by using the value of $k > 0$, find the value of a if $\lim_{x \rightarrow -k} f(x)$ exist. [5 marks]

6. (a) If $xy = 2(x-y)^2$, find the following values at $\left(1, \frac{1}{2}\right)$

(i) $\frac{dy}{dx}$ (ii) $\frac{d^2y}{dx^2}$ [9 marks]

(b) Given $x = \cos 2t$ and $y = 2t + \sin^2 t$, show that $\frac{dy}{dx} = -\operatorname{cosec} 2t - \frac{1}{2}$ and

evaluate $\frac{d^2y}{dx^2}$ when $t = \frac{\pi}{8}$. [7 marks]

7. (a) Given a water tank which has a shape of circular cone with a base radius of 3 metres and a height of 5 metres. If water being pumped into the tank at the rate of $2\text{m}^3/\text{minute}$, find the rate of change in water level rising when the water is 3 metres deep.

[5 marks]

(b) A closed cylinder of height h cm and radius r cm. The volume of the cylinder is $128\pi\text{ cm}^3$, and its total surface area is $A\text{ cm}^2$

(i) Show that $A = 2\pi\left(r^2 + \frac{128}{r}\right)$. [3 marks]

(ii) Calculate the minimum value of A . [3 marks]

END OF QUESTION

Final Answer:

SECTION A

1. (a) $-\frac{1}{2}$ (b) 2 (c) $-\frac{1}{2}$
2. (a) $3e^x \tan^2(e^x + 3) \sec^2(e^x + 3)$ (b) $-\frac{2x}{2-x^2} - \frac{1}{x} + \frac{1}{1-x}$
(c) $(1+3\ln x)x^2 2^{x^3 \ln x} \ln 2$
3. (a) $b = -9$, $c = 12$ (b) maximum point

SECTION B

1. (a) $z = 7\sqrt{65}[\cos(-2.622) + i \sin(-2.622)]$ (b) $z = -2 + i$
2. (a) $(-\infty, -3) \cup \left(-3, -\frac{9}{5}\right] \cup [3, +\infty)$ (b) $\{x : x \leq 10\}$
3. (a) $D_f = (-\infty, \infty)$; $D_g = (-\infty, -1) \cup (-1, \infty)$
(b) $fg(k) = \frac{1-5k}{k+1}$, $gf(k) = \frac{-6k-3}{2k+2}$, $k = \frac{5}{4}$
4. (a) $f(-3) = -1$ (b) D.I.Y
(c) x -intercept $(-2, 0)$, y -intercept $(0, 3)$ (d) $D_f = [-5, \infty)$; $R_f = [-3, 0] \cup [1, \infty)$
5. (a) $k = 0$, $k = 3$ (b) $a = \frac{4}{3}$
6. (a) (i) $\frac{1}{2}$ (ii) 0 (b) 2
7. (a) $\frac{dh}{dt} = \frac{50}{81\pi} \text{ m/min}$ (b) $A = 96\pi \text{ cm}^2$